

Homework - 8

1. Let L/k be a finite extension of fields, let $N \subseteq \bar{k}$ be its normal closure. Prove that $[N:k] < \infty$.
2. Let L/k be a separable extension of fields, admitting only finitely many k -embeddings in an algebraic closure $\bar{k}$ of k . Prove/disprove $[L:k] < \infty$.
3. Let $f(x) \in \mathbb{Q}[x]$ be irreducible and $L = \text{Split}(f; \mathbb{Q})$. Prove/disprove: $G = \text{Aut}(L)$ acts transitively on the set of roots of $f(x)$ in L .
4. Let α be a root of $X^3 + X^2 - 2X - 1 \in \mathbb{Q}[X]$. Verify $\beta = \alpha^2 - 2$ is also a root. Let $L = \mathbb{Q}(\alpha)$. Determine $\text{Aut}(L)$ & show that $L/\mathbb{Q}$ is normal.
5. Prove/disprove: Let L/k be a field extⁿ, $\alpha \in L$ be transcendental over k ; $f(x) \in k[x]$ be non constant. Then $f(\alpha)$ is transcendental/ k .
 - If $\beta \in L$ be an element with $f(\beta) = \alpha$, then β is transcendental/ k .
6. Construct an infinite tower (sequence) of algebraically closed fields $K_1 \subsetneq K_2 \subsetneq K_3 \subsetneq \dots$

7. Consider a field extⁿ $k(\alpha, \beta)/k$. Suppose $\alpha \notin k$, is algebraic/ k and β is transcendental/ k . Can $k(\alpha, \beta)$ be simple/ k ? Explain.

8. Prove/disprove; for a field K , the groups $(K, +)$ and $(K^*, \cdot)$ can never be isomorphic.

9. R be an integral domain containing a field k . With the natural vector space structure, suppose $\dim_k R < \infty$. Prove that R is a field.

10. Let $L/\mathbb{Q}$ be a finite extension. Prove or disprove; the torsion subgroup of $(L^*, \cdot)$ is finite.

11. Let K/k be a field extⁿ, $[K:k] = n$ & $f(x) \in k[x]$ irreducible, $m = \deg(f)$. Prove that if m & n are coprime, then f cannot have a root in K .

12. Let $f(x) = a_n x^n + \dots + a_1 x + a_0 \in \mathbb{Q}[x]$ be irreducible, $n > 1$, $a_n \neq 0$. Assume all roots of f lie on S^1 , the unit circle in $\mathbb{C}$. Prove that $a_i = a_{n-i} \forall i$.

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